Fourier Analysis: An Intuitive Journey

From "How Are Images Waves?" to Hilbert Space Forensics

CS424 Group Project — Internal Reference

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Contents

1 The Setup: What Are We Even Doing? **2**

2 Complex Numbers Live in 2D **2**

3 The DFT as a Change of Basis **2**
3.1 Why Exactly N Basis Functions? . . . 3

4 The Music Analogy (And Why It Breaks) **3**

5 2D Fourier: Images as Sums of Stripes **3**
5.1 Why a Product? . . . 3

6 Power Spectrum: Where Is the Energy? **4**

7 FFT Shift: Fixing the Array Layout **4**

8 The $1/f^{\alpha}$ Power Law **5**

9 The Functional Analysis Perspective **5**
9.1 The Hilbert Space . . . 5
9.2 The Unitary Operator . . . 5
9.3 Convolution as Operator Diagonalisation . . . 5
9.4 Steganography as Subspace Separation . . . 5

10 Connecting Everything: The Forensic Audit **6**
10.1 FFT Residual Analysis . . . 6
10.2 Perturbation Test . . . 6
10.3 The Pareto Tradeoff . . . 6

1 The Setup: What Are We Even Doing?

We want to detect whether a CycleGAN generator is "cheating" by hiding information as invisible noise in its output images. The tool we use is the Fourier Transform. But why?

The short answer: the Fourier Transform lets you look at an image from a different angle where the cheating becomes obvious.

The long answer is the rest of this document.

2 Complex Numbers Live in 2D

A complex number $z = a + bi$ is **not** a 1D scalar in the usual sense. It is a point in a 2D plane:

- Real part a = horizontal coordinate
- Imaginary part b = vertical coordinate
- $|z| = \sqrt{a^2 + b^2}$ = distance from origin

Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ maps angle θ to a point on the unit circle. As θ increases, $e^{i\theta}$ traces the circle:

$$\theta = 0 \implies (1, 0) \quad \text{(right)} \tag{1}$$
$$\theta = \pi/2 \implies (0, 1) \quad \text{(top)} \tag{2}$$
$$\theta = \pi \implies (-1, 0) \quad \text{(left)} \tag{3}$$
$$\theta = 3\pi/2 \implies (0, -1) \quad \text{(bottom)} \tag{4}$$

The function e^{ikt} spins around the circle k times as t goes $0 \to 2\pi$. Higher k = faster spin = higher frequency.

3 The DFT as a Change of Basis

A discrete signal of N samples is a vector $\mathbf{f} \in \mathbb{C}^N$. That's just an N-dimensional vector space.

Definition 3.1 (DFT Basis)**.** The N vectors

$$\mathbf{e}_k = \frac{1}{\sqrt{N}} \left(1,\ e^{-i2\pi k/N},\ e^{-i2\pi \cdot 2k/N},\ \ldots,\ e^{-i2\pi(N-1)k/N}\right) \tag{5}$$

for $k = 0, 1, \ldots, N-1$ form an **orthonormal basis** for $\mathbb{C}^N$.

The Fourier coefficient $\hat{f}[k] = \langle \mathbf{f}, \mathbf{e}_k \rangle$ is a projection—it measures how much of $\mathbf{f}$ "points in the direction" of $\mathbf{e}_k$. This is identical to how $\langle \mathbf{v}, \mathbf{e}_1 \rangle$ gives the x-component of a 3D vector.

Key Insight

The DFT is nothing more than a matrix multiplication $\hat{\mathbf{f}} = \mathbf{W}\mathbf{f}$ where $\mathbf{W}$ is unitary. It is a **rotation** (more precisely, a unitary change of basis) in $\mathbb{C}^N$. No information is lost. You are looking at the same vector from a different coordinate system.

3.1 Why Exactly N Basis Functions?

If your signal has N samples, your vector space is N-dimensional. By the fundamental theorem of linear algebra, any basis for this space has exactly N vectors—no more, no fewer.

- Fewer than N: cannot reach every point in the space (reconstruction fails)
- More than N: the extra vectors are linearly dependent (redundant)

The basis function $e^{-i2\pi \cdot N \cdot n/N} = e^{-i2\pi n} = 1$ for all integer n. So $k = N$ produces the exact same vector as $k = 0$. This is **aliasing**—frequencies beyond $N - 1$ wrap around and repeat.

4 The Music Analogy (And Why It Breaks)

Common Misconception

"A 3-note chord needs 3 basis functions, one per note." **Wrong.**
A musical note is **not** a single complex exponential. A real piano note has a fundamental frequency plus dozens of overtones, transients, and decay. Reconstructing one note requires non-zero coefficients for hundreds of DFT basis functions.
Even a pure sine wave requires **two** basis functions (positive and negative frequency) to produce a real-valued signal, due to conjugate symmetry.
The DFT basis functions are the "atoms." A note is a molecule built from many atoms.
N is determined by your sample count, not by how many notes you play.

5 2D Fourier: Images as Sums of Stripes

An image $f[m, n]$ of size $M \times N$ lives in $\mathbb{C}^{M \times N}$, which has MN dimensions. The 2D basis functions are tensor products of 1D basis functions:

$$e_{u,v}[m, n] = e^{-i2\pi um/M} \cdot e^{-i2\pi vn/N} = e^{-i2\pi(um/M + vn/N)} \tag{6}$$

- u controls oscillation speed along the horizontal axis
- v controls oscillation speed along the vertical axis
- Together, they produce **stripes** at angle $\arctan(v/u)$ with frequency $\sqrt{u^2 + v^2}$

5.1 Why a Product?

It is not a coincidence. If $\{\phi_u\}$ and $\{\psi_v\}$ are orthonormal bases for $\mathbb{C}^M$ and $\mathbb{C}^N$ respectively, then $\{\phi_u \cdot \psi_v\}$ is an orthonormal basis for $\mathbb{C}^{M \times N}$.

Proof sketch: The inner product factorises because the sums over m and n are independent:

$$\langle e_{u,v}, e_{x,y} \rangle = \underbrace{\left(\sum_m \phi_u[m] \overline{\phi_x[m]} \right)}_{\delta_{u,x}} \cdot \underbrace{\left(\sum_n \psi_v[n] \overline{\psi_y[n]} \right)}_{\delta_{v,y}} = \delta_{u,x} \delta_{v,y} \tag{7}$$

Zero unless both indices match. This extends trivially to 3D, 4D, etc.

Common Misconception

"Can I just flatten the image to 1D and use a single index $k \in [0, MN-1]$?"
Mathematically: yes, $\mathbb{C}^{M\times N} \cong \mathbb{C}^{MN}$. Practically: **catastrophic.** Flattening destroys the 2D spatial topology. Pixel $(0, N-1)$ at the end of row 0 and pixel $(1, 0)$ at the start of row 1 are spatial neighbours but become distant in the flattened array. The 1D basis functions cannot represent 2D features (edges, textures) cleanly.
Use (u, v) because (x, y) exists. Respect the geometry.

6 Power Spectrum: Where Is the Energy?

Definition 6.1 (Power Spectrum)**.**

$$P[u, v] = |\hat{f}[u, v]|^2 = \mathrm{Re}(\hat{f}[u, v])^2 + \mathrm{Im}(\hat{f}[u, v])^2 \tag{8}$$

This measures the **energy** (variance) at frequency (u, v). By Parseval's theorem, total spatial energy equals total spectral energy:

$$\sum_{m,n} |f[m, n]|^2 = \frac{1}{MN} \sum_{u,v} |\hat{f}[u, v]|^2 \tag{9}$$

Common Misconception

"The power spectrum measures the magnitude of the basis function." **Wrong.** The magnitude of $e^{i\theta}$ is always exactly 1. The power spectrum measures the magnitude squared of the **coefficients**—how much weight each basis function carries in the decomposition of your specific image.

7 FFT Shift: Fixing the Array Layout

The raw DFT output stores frequencies in a counterintuitive order:

- Index 0: DC component (zero frequency)
- Indices 1 to $N/2$: positive frequencies (increasing)
- Indices $N/2 + 1$ to $N - 1$: **negative** frequencies (due to periodicity)

The key identity: $e^{-i2\pi(N-1)n/N} = e^{+i2\pi n/N}$. Index $N - 1$ is frequency -1, not a high frequency. It has **nearly the same energy** as index 1.

Without shifting, the bright low-frequency energy is split across all four corners of the 2D array. `fftshift` swaps quadrants diagonally, placing the DC component at the center. After shifting:

- **Center** = zero frequency (brightest)
- **Edges** = Nyquist frequency (darkest for natural images)
- Energy decays radially outward

This is required for correct radial averaging—without it, your distance-from-origin calculation is geometrically wrong.

8 The $1/f^\alpha$ Power Law

Natural images obey a statistical law: their radial power spectrum decays as

$$P(\rho) \propto \frac{1}{\rho^\alpha}, \qquad \alpha \approx 2 \tag{10}$$

Why? Physical objects have smooth surfaces. Adjacent pixels are highly correlated. Smooth regions are low-frequency objects. Sharp edges exist but are sparse.

Key Insight

This power law means natural images are **sparse** in the Fourier basis. Nearly 99% of the energy concentrates in <1% of the basis dimensions (the low frequencies). The high-frequency subspace is practically **empty**.
This is the mathematical real estate where steganographic data hides. The power law guarantees the space is empty; the Fourier transform gives you the map to inspect it.

9 The Functional Analysis Perspective

Translating everything into abstract algebra:

9.1 The Hilbert Space

An $M \times N$ image is a vector f in the finite-dimensional Hilbert space $\mathcal{H} = \mathbb{C}^{MN}$, equipped with the Hermitian inner product:

$$\langle f, g \rangle = \sum_{m,n} f[m,n]\overline{g[m,n]} \tag{11}$$

9.2 The Unitary Operator

The DFT is a unitary operator $U : \mathcal{H} \to \mathcal{H}$ satisfying $U^*U = I$. It is an **isometry**: it preserves inner products, norms, and angles. The Fourier and pixel representations contain identical information, viewed from different bases.

9.3 Convolution as Operator Diagonalisation

Convolution with kernel g is a linear operator (a circulant matrix C_g). The Fourier basis **simultaneously diagonalises all circulant matrices**:

$$C_g = U^* \Lambda_g U \tag{12}$$

where Λ_g is diagonal with entries $\hat{g}[k]$. This is why the Convolution Theorem works: convolution becomes pointwise multiplication in the eigenbasis.

9.4 Steganography as Subspace Separation

Due to the $1/f^\alpha$ law, natural images occupy a low-dimensional subspace of $\mathcal{H}$: their projections onto high-frequency basis vectors are near zero. Steganographic noise is pseudo-random with uniform spectral energy.

In the spatial basis, signal and noise are entangled at every pixel. In the Fourier basis, they naturally separate:

- Low-frequency subspace: $\|\langle f_{\text{cover}}, e_{\text{low}} \rangle\| \gg \|\langle w, e_{\text{low}} \rangle\|$ (signal dominates)

- High-frequency subspace: $\langle f_{\text{stego}}, e_{\text{high}} \rangle \approx \langle w, e_{\text{high}} \rangle$ (only noise remains)

Connection to FB-CycleGAN

The Gaussian blur in FB-CycleGAN is a circulant operator. In the Fourier basis, it becomes diagonal: $\Lambda_B[k] = e^{-2\pi^2\sigma^2k^2}$. This zeros out exactly the high-frequency subspace where steganographic data lives, while preserving the low-frequency structural content. The generator can no longer write to those coordinates because the blur operator annihilates them before the reverse generator reads them.

10 Connecting Everything: The Forensic Audit

Now we can state precisely what the forensic audit measures:

10.1 FFT Residual Analysis

Compute the residual $r = |G(x) - x|$ and project it into the Fourier basis. In an honest translation, r is spatially localised (only the tumor region changes), which means its energy concentrates at low frequencies (the $1/f^\alpha$ law). In a steganographic translation, r contains hidden data spread across all pixels, which means its high-frequency coefficients are anomalously large—the power spectrum plateaus instead of decaying.

10.2 Perturbation Test

Add noise $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ to the generated image. White noise has uniform spectral energy. At high frequencies where the hidden signal is faint, even tiny noise overwhelms the steganographic coefficients (SNR $\to 1$). The reverse generator can no longer read the hidden message, and reconstruction collapses catastrophically.

An honest generator reconstructs from low-frequency structural features, which are robust to noise (the signal is orders of magnitude larger than the perturbation in that subspace). Reconstruction degrades gracefully.

10.3 The Pareto Tradeoff

Increasing σ in the Gaussian blur lowers the cutoff frequency, shrinking the subspace available to the generator. At some σ^*, the available subspace is too small to encode the tumor coordinates, forcing honest translation. But too large σ also destroys legitimate high-frequency detail (edges, texture), reducing translation quality. The Pareto analysis sweeps σ to find the optimal operating point.